

Team Control Number: 12263, Problem Chosen: C, 2021 MidMCM

1. Summary

The problem that we chose was Problem C. This problem requires us to create a dog park complete with justifications for every decision we make. This dog park needs to be within the border of a certain region, provided with a scale drawing map. The maximum area for this dog park can be one acre or 43,560 square feet. The problem also states that during the process of designing our dog park, we need to leave the skate park in its existing form. We do not have a cost limit, although all our purchases should be reasonable. The problem also provides information about different dog park locations and what visitors like in their dog park which are but are not limited to easy access to clean water, a place to comfortably sit and/or relax, and a large area to walk and play.

In the planning section, the question asks us to make a list of necessary items needed to build our park. It also asks us to make a list of questions and assumptions to make necessary decisions about our dog park. Next, in the designing section, asks us to make scale models of our park and bill-of-materials. Bill-of-materials lists the items we will buy for our park as well as how expensive each item is, how many we will buy, and the total cost of all the items combined. Additionally, we had to make a list of other expenses, ones that we didn't include in the BOM. The problem also wants us to consider our variables. For example, if the Middleville Director decided to provide us with more or less amount of money what would we do? What about our space? Finally, we had to share our solution. Writing a letter to the Director, creating a summary, a reference list, and a table of contents is what this section includes.

Our solution includes where we put our dog park, what we put in our dog park, and all our justifications. We located our dog park near parking, so visitors didn't have to walk a long way to get to our park, and away from the skate park, so dogs didn't get distracted or scared by the skaters. We then figured out the shape we wanted our dog park to be. We knew we wanted it to have minimal corners and for it to be easy to split up. We also wanted a maximum area with little fencing. We needed these things because we require different spaces for small dogs, all dogs, and the vending machine area.

After considering these points, we chose a composite shape. This shape consists of a square and two half circles. Our next step was to find the area to make sure it meets the criteria of our problem. Our equation to do this is $\pi r^2 + 4r^2 = A$. After we solved for our maximum area, 43,560 square ft, we got our radius, 78 ft. In addition to this, our park has many design features that set us apart from other parks like a dog treat vending machine and much more. These combined points are how we know our park meets the criteria.



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2. Introduction

2.1 Background

According to the American Veterinary Medical Association, the percentile of dogs in households is 38.4% based on the 2017 to 2018 U.S. Pet Ownership survey.¹ A majority of people need a space to play with their dogs. The solution to this problem is a dog park. Dog Parks became popular in 2009, 466, and increased to 774 dog parks in 2018 according to, "The Trust for Public Land."² This leads us to the point of needing more quality dog parks for the common public. This is true because of the fact, not all dog owners have space in their yards for their dogs to actually enjoy themselves. To further evaluate, the average house in the Suburbs has a lot size of one-fifth of an acre³. For a dog to get the most out of its outside time, the dog needs enough open space where they can run freely. This is where dog parks come in. Dog parks are places where you can go with your canine friend and enjoy yourselves. While you are outside you can take your dog off-leash, play fetch, or just sit and watch the clouds drift by. All dog parks are large open spaces enclosed by a tall fence to stop dogs from sprinting off into dangerous lands.

2.2 - Problem Interpretation

In this assignment, we have been asked to make an authentic design of a dog park with a maximum space of one acre, while not touching the forests and the skate park. We first approached by finding a shape for our dog park that we can get the most out of, without putting anyone in close proximity to the dangerous roads. After that, we found all the necessary features needed for a successful dog park. To finish the thinking process we found items that would improve the quality of life for dogs and humans alike.

3. Inquiry

3.1 - Necessary Features

There are many different features a dog park needs and many features we would like to have in a dog park. These needed features are necessities that keep the dogs and humans occupying the park safely. These include water, shade, food, seating, and a large open space. There are other elements that, while fun or nice to have, are not necessary for a successful park. Here is a list of said additions:

- Dog showers
- Picnic tables
- Plastic bag dispensers
- Materials for dog amenities



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The reason we have considered these items secondary is that the dogs and humans using this park can easily live without them. With the dog showers, a dog owner could just bring a towel to clean off their dog after they are done using the park, in terms of the picnic tables, you might need a bench to sit but you do not need a full table to do that, With the plastic bag dispensers, the owner could and should bring their own doggy bags to clean up after their dog, and with the materials for dog amenities, the amenities themselves are not necessary so the materials for them are not necessary. There are also some non-items that are necessary, like rules. To ensure that everyone has the best experience that they can, we need some regulations. One that we have is in the concessions area, people need to put their dog on a leash so that dogs do not disturb other visitors while they are eating. In addition, they need to clean up after their dogs.

3.2 – Questions

There are still many questions to answer about our dog park design. Many of these questions are questions we have asked ourselves. We asked these questions so we could figure out what we needed to work on. Such questions are,

- What shape and what size will our dog park be?
- Where in the available area will we locate our dog park?
- What materials will we need to build the park?
- How much will it cost to build and maintain the dog park?
- Will the park be free to use?
- What is necessary for a dog park?
- How big will the park be?
- What features will our park have that sets it across from other parks
- How will we make a profit?
- What hours will the dog park be open?
- What are the safety guidelines for a dog park?
- Do we need to fence in the concession area?
- Why should you pick our model?
- What goes into the maintenance of the park?
- What do we have in terms of shade?
- How can we calculate the price of all of the items?
- How do we keep dogs from running out of the main gate?

This list of questions is for a very simple reason. This reason is it helps us make certain choices and conclusions in the design by giving us a footing on certain parts of the thinking process. These footings include being able to realize new and better ideas, weed out the bad or unreasonable thoughts, and overall create more effective strategies within the group.



3.3 – Assumptions

We have also made a list of assumptions and justifications that have helped us answer some of these questions.

Assumption 1: Big dogs need more room than little dogs.

Justification 1: Big dogs have longer legs, taller bodies, and faster running speeds, therefore, meaning they need more room to have the same amount of enjoyment as the small dogs.

Assumption 2: The Park needs open and closed times.

Justification 2: If a dog park did not have an open and closed time then we would have to hire staff willing to work twenty-four hours a day or hire workers willing to work in the very middle of the night. If we don't do this our vending machines and other valuable assets to the park could be stolen.

Assumptions 3: Dogs need some kind of cleaning.

Justification 3: After a rainy day, the dirt that is used under the play area's grass will be muddy. If a persons' dog attempts to play in said area that dog will become messy as well. If this dog is not properly cleaned before going back home, the owner's car will become a mud-scape itself.

Assumption 4: Owners want to have quick access to dog treats.

Justification 4: Studies show “A majority (58%) of pet owners pamper their pets by giving them indulgent treats”.⁴ This shows us that a majority of dog owners wish to have access to treats to “pamper” their dog with.

With these assumptions, we created a list of variables that would help us design our dog park. See Table 1 for our complete list of variables.

Table 1 - Variable Information

VARIABLE NAME	VARIABLE SYMBOL
Enclosed Area of the Whole Dog Park in Square Feet	A
Enclosed Area of Small Dogs Park in Square Feet	X
Enclosed Area of All Dogs Park in Square Feet	B
Unit Price of Each Material in Dollars	a, b, c,...(assigned to each item)



Overall Cost in Dollars	C
Number of Materials for Each Item	m_i (i is item number)
The Exterior Perimeter of Small Dogs Section in Feet	S
Small Dogs Section Complete Perimeter	T
All Dogs Section Complete Perimeter	V
The Exterior Perimeter of All Dogs Section in Feet	L
The Exterior Perimeter of Whole Dog Park in Feet	P
The Radius of Semicircle in Feet	r
The Side Length of Square in Feet	2r

4. Developing Our Models

4.1 - Consideration of Shapes

In this section, we considered different shapes for our park, shown in Table 2. We wanted to simplify the given area to a much simpler version so that it was easier to compare each shape. That is why we explored different shapes with a fixed area of 400 square feet. We found the lengths, widths, and radius for the areas of shapes given in Table 2. We then found the perimeter to see how much fencing we might need around the park. We kept the area the same so we could have a control variable and because we have a given area. We wanted to find the shape with the least perimeter so we could have less fencing to buy. A pattern we saw was if the side measurements were closer together then the perimeter was smaller.

Table 2 - Shape Analysis

SHAPE	AREA (sq ft)	LENGTH (ft)	WIDTH (ft)	PERIMETER (ft)
Square	400	20	20	80
Rectangle 1	400	10	40	100
Rectangle 2	400	4	100	208



Rectangle 3	400	25	16	82
Regular Hexagon	400	~12.41	~12.41	~74.45
Circle	400	$r = \sim 11.3$	$r = \sim 11.3$	~71

Initially, we considered different shapes by trial and error. We first considered a square. We liked this idea because it was easy to split up and work with, and we could find the side lengths very easily. Something we disliked, though, was the amount of fencing we needed to buy. So, we considered this shape but moved on. We then considered a rectangle, which was very similar to a square. It was still easy to work with, but the perimeter was even larger. Then we considered a hexagon. This was very hard to split up and find the area of, even though the perimeter was low. Finally, we came up with the idea of a circle. This was hard to split up, but the perimeter was the lowest out of all of them.

We found the measurements in Table 2 by plugging in our area and finding factors, or our length and width. We then found the perimeter using these lengths and widths, so we could find what had the most area for the least amount of fencing. Here are sample calculations:

Sample 1. Analysis of a square with an area of 400 sq. ft

$$\text{Area} = 400 \text{ sq. ft} = s^2 \text{ (} s \text{ representing measurement of one side)}$$

$$s = \sqrt{400} = 20 \text{ ft}$$

$$\text{perimeter} = 4s = 4 \times 20 = 80 \text{ ft}$$

Sample 2. Analysis of a circle with an area of 400 sq. ft

$$\text{Area} = 400 \text{ sq. ft} = \pi r^2 \text{ (} r \text{ representing radius of the circle)}$$

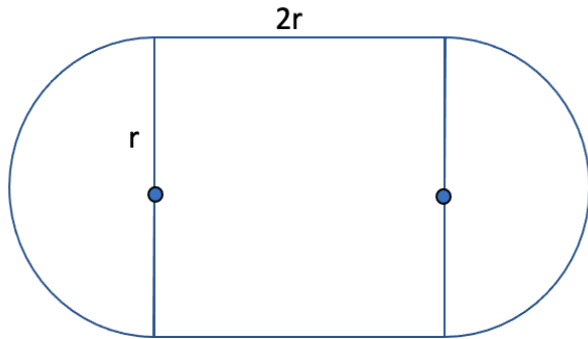
$$r = \sqrt{(400 \div \pi)} \cong 11.3 \text{ ft}$$

$$\text{perimeter} = 2\pi r \cong 71 \text{ ft}$$

After we considered all the options in Table 2, we settled on a composite shape. This shape was composed of two shapes from Table 2, a square and a circle. For a given area these shapes had reasonable fencing considerations. See Figure 1 for a sketch of this composite shape.



Figure 1 - Composite shape for the dog park

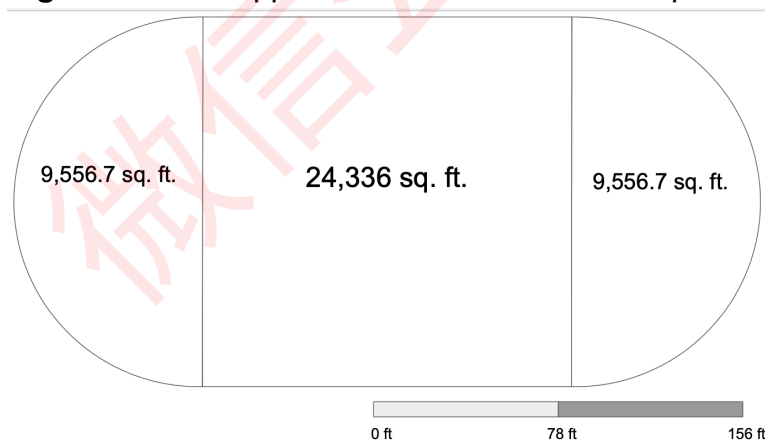


This composite shape made from two semi circles and a square. To measure the areas of these shapes, though, we had to find the side lengths. We did this by using our equation

$$A = \pi r^2 + 4r^2$$

In this equation the variable r represents radius, and we plugged in our maximum area, 43,560 ft. in for A . When we solved for r we got 78.09916... We rounded this number down to the closest whole number, so we got 78 ft for our radius. We used underestimation because we needed our region to be less than or equal to the maximum space provided. This showed us the radius of our 2 half circles. Then we needed our diameter or $2r$. This showed us the length and width of the square, and the height, or diameter, of our half circles. We then calculated the areas as shown in Figure 2. These areas added up to approximately equal to 43,449 square ft., which is right under our maximum area, 43,560 square ft.

Figure 2 – The Approximate Area of Each Shape



4.2 - Justification of Design

Once we had the outline of the dog park we worked on our interior design. We wanted to divide our dog park into three sections. We figured out the separation because we wanted a space for small dogs, concessions, and all dogs. We didn't put in a space for only big dogs because they are not in danger to be with the small and medium-sized dogs, while small dogs are. We gave the small dogs section a half circle and $\frac{1}{3}$ of the main square. This is for dogs under 35 pounds.⁵ This was because we needed a lot of space for small dogs because owners in the U.S. have more small dogs than big dogs.

According to the Humane Society of the United States, a higher percentage of US households have small dogs (under 25 pounds) than medium dogs (25-40 pounds) or large dogs (40+ pounds), with the figures at 48%, 33%, and 33%, respectively.⁶ We used this information to find a ratio between small dogs and other dogs. The ratio was 48 to 66, we simplified and got 8 to 11 and rounded up to 8:12.

$$8:12 = 2:3$$

We totaled how many dogs there were and then looked at the relationship between small dogs, $\frac{2}{5}$, out of all dogs, and other dogs out of all dogs, $\frac{3}{5}$. The sum of the percentages in the survey does not add up to 100 percent, but we just assumed that people might have multiple dogs in their households. To capture this percentage distribution, we decided to give one of the semicircles and one-third of the square. When we computed the area of the small dogs, we found we took up around 40% of the space.

$$\frac{2}{5} = 40\%$$

$$X = \frac{\pi * 78^2}{2} + 156 * 52 \cong 17,669 \text{ sq. ft.}$$

$$\frac{\text{Small Dogs Area}}{\text{The whole Area}} = \frac{X}{A} = \frac{17,669}{43,449} \cong 0.41 \cong 41\% =$$

We are left with the two-thirds of the section of the square and the other semicircle as the remaining area. We then sectioned off one-third of the remaining square and used that as concessions. This was because we needed to fit restrooms, vending machines, and picnic tables. The rest was left for all dogs' area. You can see this in Figure 3.

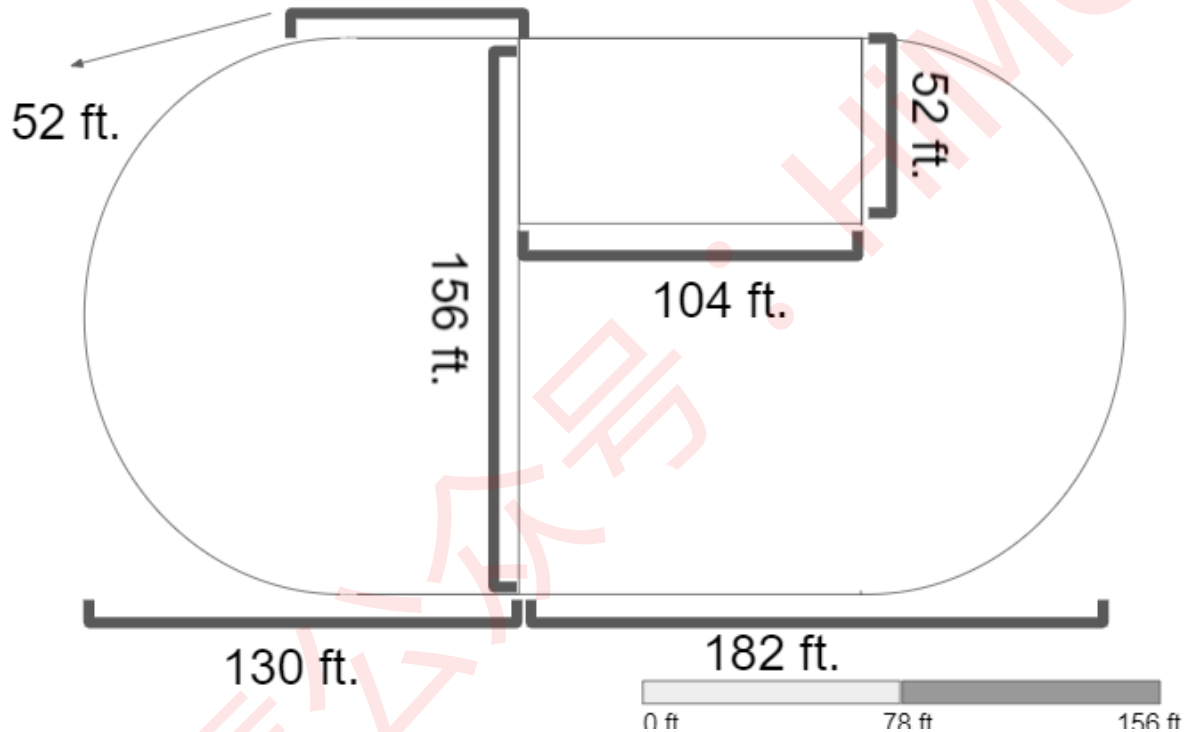


$$B = \frac{\pi * 78^2}{2} + 104 * 104 \cong 20,372.73 \text{ sq. ft.}$$

$$\frac{\text{All Dogs Area}}{\text{The whole Area}} = \frac{B}{A} = \frac{20,374}{43,449} \cong 0.47 \cong 47\%$$

All-dogs area took up about 47% of the area. and the concessions, restroom-related spaces took up around 12% of the area.

Figure 3 - Scale Drawing Analysis



We calculated the total length from one end of the park to the other for each dog park. We wanted to see if the dogs would have enough space to enjoy themselves playing fetch, running, etc. The measurements we ended up with a length of 130 ft. for the small dog area, and 182 ft. for the all-dogs area. Pet playgrounds states, "For most breeds we recommend a 5 foot tall fence and a minimum of 200 linear feet of fencing".⁷ Small dog perimeter is about 505.04 ft which is more than enough feet for a dog to run around in. The all-dog perimeter is about 609.04 ft which is also more than enough space for dogs to run around.

$$T = \pi * 78 + 52 * 2 + 156 \cong 505.04 \text{ ft}$$

$$V = \pi * 78 + 52 + 104 * 3 \cong 609.04 \text{ ft}$$



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4.3 - Scale Drawing with Area

Our justification for putting our dog park where we did is that it is close to parking and far away from the skate park. Dogs can easily get distracted, and we don't want them to get scared of the skaters. We also want our park to be close to parking, so our visitors don't have to walk a long way with their dog. We labeled four sections on the map provided (See *Figure 4* for all sections).

Figure 4 - Park Measurement Analysis



We chose Section 1 because it was the greatest section out of all of them. We eliminated Section 3 because it had lots of trees and we want to preserve nature. If we chose that area, we would have to cut down a great number of trees. Section 4 did not have the space we needed for our dog park, so we did not choose that one either. Section 2 was not near the parking compared to Section 1. Finally, we looked at Section 1 and saw that it was near parking and was away from the skateboarding park if you angled our park right.

We know the area of our dog park fits into section 1 because we measured the area that we must fit our dog park in. Using scale measurement provided in the map, we measured each sections' perimeter in their specified border. We then measured our park to scale, to make sure it fit and inserted the scale drawing into the park. We calculated the available section 1 area, the top left section separated by the two roads.



To calculate the area, we approximated the width and length including the skateboard park, 300 ft. Next, we found the height by using the Pythagorean Theorem which we got around 223.61 ft.

*The whole section 1 area $\cong 223.61$ (height) * 300 (base) $\cong 67,083$ ft*

*Skateboarding park area $\cong 100 * 100 \cong 10,000$ ft*

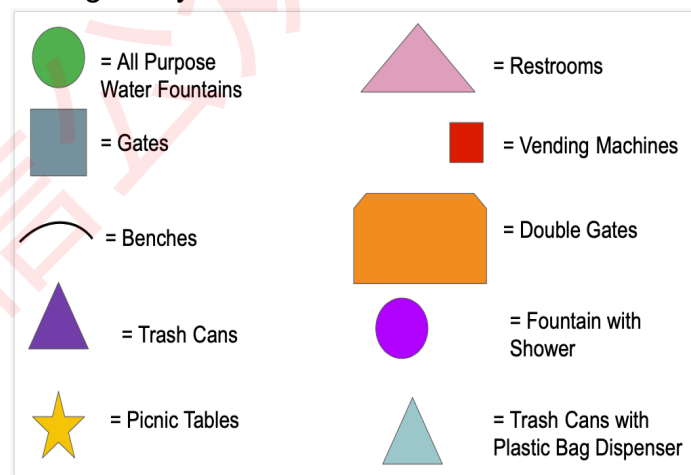
Available section 1 area $\cong 67,083 - 10,000 \cong 57,083$

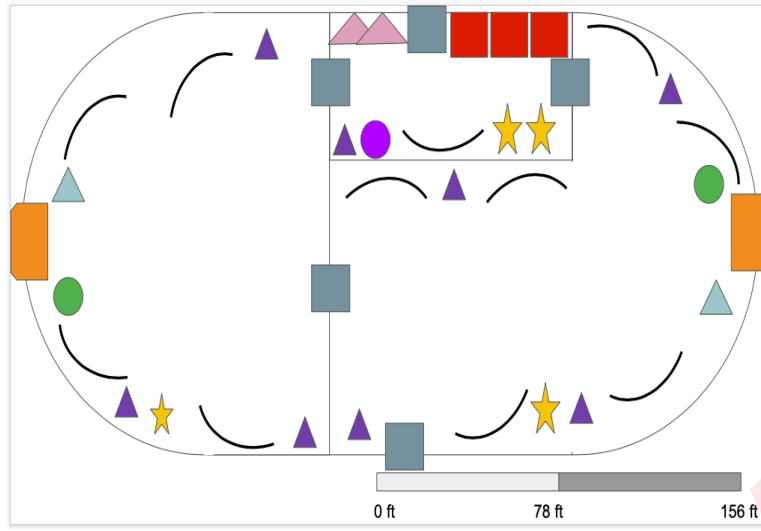
The dog park area (43,444 sq ft) is smaller than 57,083 so our proposed dog park will fit into section 1.

4.4 - Scale Drawing and Key Explanation

This key and scale drawing image in Figure 5 shows us an object and the amount of that object that we have. We will talk about how we placed these items later on. We will also include signs, like to put your dog on leash and clean up after your dog. Since this is a public space, we consider having fines and regulations in place so that our dog park stays clean. According to the Animal Law website parks can request a fine to enforce cleaning up after their dogs.⁸ These fines would be determined in collaboration with Middleville parks and recreations office. A list of general rules will be displayed in the concession area. In addition to these, we also included features like concrete and picnic tables in the concessions stand. This is so that people don't have to be worried about getting dirty while eating.

Figure 5 - Scale Drawing Analysis with Amenities





As you can see in the design features, we have a concession area with two picnic tables and 3 vending machines. One of the machines is for food like chips and sandwiches, another is for dog treats, and the final one is for drinks. Dog owners can use this space to socialize with their friends as they pamper their dogs with some treats. Dogs are expected to be on leash in this area. The area we chose to build our dog park has some trees and we hope to keep those trees because they would provide shade for dogs and dog owners. We also plan to have more trees spread around the park, keeping the open space.

4.5 – Bill of Materials

We created our Bill of Materials by looking at each item's description and size and found how much of each amount we needed. See Table 3 for a complete list of the bill of materials. We found the unit price from online stores, such as the Home Depot, Amazon, etc., and multiplied the amount needed by the cost per item to find the total cost. You can find the links for these items in the references section. We also classified the items into categories of negotiable and non-negotiable. Negotiable means that we can change the items or even remove them. Non-negotiable means that the items are needed, and we will try to add more of them. By using our scale drawing, we figured out the items we wanted to get, how many of each item, the cost, and if they were negotiable or non-negotiable.

Table 3 - Bill of Materials for Dog Park

ITEM CATEGORY	ITEM NUMBER	DESCRIPTION AND SIZE	AMOUNT NEEDED	COST PER ITEM	TOTAL COST	VARIABLE ASSIGNED



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Non-Negotiable	1	Trash Cans 33gal	8	54	432	a
Non-Negotiable	2	Plastic Bag Dispense r with Trash Can 73in x 18in x 10in	2	325	650	b
Negotiable	3	Fountain with Shower	1	3,151	3,151	c
Non-Negotiable	4	Fence 5ft x 50ft	22	596	13,112	d
Non-Negotiable	5	All-purpose Fountain w/ two pet bowls	2	3,980	7,960	e
Non-Negotiable	4	Gate 6ft x 4 ft	5	229.99	1,149.95	f
Negotiable	7	Picnic Table 46in x 18in x 46in	4	1,670	6,680	g
Non-Negotiable	8	Benches	11	735	8,085	h
Negotiable	9	2 Pressure Treated Timber 4in x 4in x 8ft	3	11.28	33.84	i
Negotiable	10	Edge Glued	6	9.09	54.54	j



		Rounds 1in x 17- ¾ in				
Negotiable	11	Dog Walk Ramp (All dog area)	1	1,970	1,970	k
Non-Negotiable	12	Double Gate Chain Link Fence 6ft x 10ft	2	309	618	l
Non-Negotiable	13	Vending Machines	3	2,195	6,585	o

The final cost we got after adding all the total costs of each item is \$50,481.33. The final cost we got after adding all the non-negotiable items is \$38,591.95. The final cost we got after adding all the negotiable items is \$11,889.38. We found all of these by using the equation

$$m_1 * a + m_2 * a + m_3 * c + \dots = C$$

The m_i in the formula represents the number needed for each item where i changes according to the item number on our list. Because each of the items in the BOM has a corresponding variable, you can replace/add/subtract different items that you want to incorporate, such as the negotiables.

4.6 - Justifications for Items and Costs

We need 8 trash cans and 2 with plastic bag dispensers because we needed to place them in the park evenly spaced so that people wouldn't throw their trash on the ground. The equation we used for the outside perimeter of the small dog area (S) is

$$S = \pi * 78 + 52 * 2$$

$$S \cong 349.04 \text{ ft}$$

We got this equation by finding the circumference of the half-circle, adding to $\frac{1}{3}$ of the square's perimeter's two lengths, to find the perimeter of the "small dogs" section. We came up with this equation by finding the circumference of the half-circle, along with the



perimeter of each side in that area except for the one connecting the all-dog area with the small dog area. Then we divided it by 4 because there are 4 trash cans in this area, including one with a plastic bag dispenser.

$$S \div 4 = 87.26 \text{ ft}$$

So, the approximate distance between each trash can is about 87.26 ft. We say this is approximate because each trash can take up about 12 to 18 inches.

Similarly, we then found the amount of space between the trash cans in the all-dog fencing using the equation

$$L = 52 + 104 * 2 + 78 * \pi$$
$$L \cong 505.04 \text{ ft}$$

We came up with this equation by finding the circumference of the half-circle, along with the perimeter of each side in that area except for the one connecting the all-dog area with the small dog area. We didn't use this side because we didn't want anything interfering with the gate. Then we divided it by 5 because that is how many trash cans we placed in that area. One of these trash cans is one with a plastic bag dispenser.

$$L \div 5 \cong 101.01 \text{ ft}$$

So, the approximate distance between each trash can is about 101 ft. We say this is approximate because each trash can take up about 12 to 18 inches.

We need the fountain and shower because most dog owners don't want their dogs to go into their house dirty, so we can have a fountain to clean their dogs, so the dogs come home nice and clean. We thought one should be enough considering not all of the dogs come and leave at the same time. We want double doors because they decrease the chance of dogs running away. According to the Association of Professional Dog Trainers, "Parks with a two-gate system avoid the possibility of dogs escaping from the park, increasing the safety of all the dogs."⁹ We added two double-gates, one to the small dog area and one to the all dog area.

We figured out the cost of the fencing, and how much we would need. To do this, we found the perimeter using the equation.

$$P = 2\pi r + 2.2r$$

In this context, $2r$ corresponds to 156 ft. This is the diameter of the circle and the length and width of the main square. We found the exterior perimeter by using 78 ft for r .

$$P = 2\pi * 78 + 2.2 * 78$$

$$P \cong 489.84 + 312 \cong 801.84 \text{ ft}$$



We then needed to find the interior perimeter, by finding two-thirds of $2r$, which is 140. Then we needed to find one-third of $2r$, which is 52, plus $2r$, or 156. All of these lengths added together are 312 ft. The corresponding length measurements are shown in Figure 3. Added to the exterior perimeter, our overall perimeter is 1,113.84 ft. After that, we found the fencing and gates we needed. These are needed because people need to exit and enter our park. First, we found the gates, which are 4 ft. wide. We got 6 of those, so we had to multiply. This equals 24. We also needed to find the double gates' widths. The double gates we're getting are also necessary because people with equipment for patience need to be able to enter without problems. These gates are 10 ft. wide, and we were getting 2 of these, so this would be 20 ft. altogether. Then we had to add the regular gate width and the double gate width together, which was 44 ft. We then needed to subtract this from the perimeter, which is the first step to finding the number of fencing we needed.

$$\text{length of fencing needed for the park} = 1113.84 - 44 = 1069.84$$

$$1069.84 \div 50 = 21.3968$$

We got a decimal number, and because of this, we had to round up, since we did not want an open space in our park's border, and we could not order a decimal of a package. So, we needed 22 fencing items. When it comes to our concession area, we will need fencing around it. You wouldn't want dogs to suddenly jump up onto random people trying to eat. This very reason is why we will have a rule where all dogs will have to be on leash in the concession area.

The reason for where we put the features where they were is that for the benches, we wanted it to be a walkable distance away and not too far away from one bench to another. The benches are around 87.23 ft apart in the small dog area and are 84.153 ft apart in the all-dog area. To find this we calculated the exterior perimeter without the interior gates in each section, the small dog section's perimeter is 348.92 ft, and the all-dog section is 504.92 ft.

$$348.92 \div 4 = 87.23$$

$$504.92 \div 6 = 84.153$$

We divided the perimeters by how much benches there is per section.

We placed our picnic tables in the place where the dogs relax because we wanted to have a place to eat in case someone wants to eat while their dog plays. We also placed them in the concessions area so people could have places to sit down and eat without fear of a dog coming and snatching their snacks.

We have decided on using vending machines instead of a concession stand so we do not have to rely on volunteers or staff to give out food. We wanted three vending machines so everyone, humans, and dogs, can eat and drink here. One of them for



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drinks, another for dog treats, and a final one for snacks. The food and drink we fill our vending machines with will be considered as an additional cost because of the fact we have to refill the machines on a regular basis.

We need the timber and the rounds so we can build dog step stools for the dogs to play on which goes hand in hand with the ramp, which helps us build a small size obstacle course. We got 3 of the timber because these came in packs of two, and we wanted 6 stepping stools. This is also why we got 6 rounds.

4.7 - Additional Considerations

We have additional considerations for building the park. We will need a water supply for the restrooms, water fountains, and the dog shower. We also would need electricity for the vending machines and for the restrooms. We would need to build the restrooms. We also would need to pour the concrete on the ground for the concessions area. We will need all the construction material, like hammer and nails, to build some items. Finally, we would need the workers to build and install everything. Finally, we will need signs such as, stop sign on the road, and just signs to put our rules on like the putting your dog on a leash in the concession area.

Maintaining the park will also need some additional considerations. We would need janitors to clean the restrooms and to take the trash out so the trash service can get it. But for the janitors to bring out the trash we would need the trash service to come and pick up the trash. In addition, we need a lawnmower service to mow the grass. The ground for the playing area in the dog park will be completely grassed so the dogs have a soft area to run and lie down. The ground for the concession area, however, will be pavement. This will be made concrete for ease of cleaning and maintenance. We came to the decision on these choices by thinking about the best interests of the people and dogs alike. We realized dogs have sensitive pads on their paws meaning they cannot run on hard surfaces for long without causing serious damage. People, though, are most likely wearing shoes so the concession stand is free to be a hard surface giving the users of the park a solid, dry surface to stand, congregate, and eat without fear of getting messy from the mud.

4.8 - Generalization Considerations

If Middleville were to have more or less space, we would get the amount of area we had and plug it in for A (A represents area and r represents the radius) in the equation

$$\pi r^2 + 4r^2 = A$$
$$r = \sqrt{\frac{A}{(\pi + 4)}}$$



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If you solved the equation, for the case of $A = 400$, r would equal around 7.48. Another way to find the dimensions of the new region would be to use a proportional reasoning approach. We can proportionally increase or decrease each section's area and the dimensions of the region would scale up or down by the square root of the scale factor. This shows that when you proportionally decrease the area, the length and the width also decrease, and the same works with increasing. A simple example of this would be if you scale up your area by 4 the dimensions will approximately increase by 2. If the area of our region was 400 and if we scaled up this region by 4 the new area would be 1600. Then the radius would be approximately 2×7.48 which is 14.97. If we plug in the new r value to our area formula the new area would be 1600,08 square feet.

You can calculate the number of items by figuring out how much space you have, what you are using that space for, and how big the items are. For example, if you have small benches, you are going to want more of them than if you had bigger benches. If your area was a concession stand, you would want more picnic tables than if your area was a play area. If it was a big area, you would want more benches than if it was a small area. Once you figure out how many of one object you want, you find it on a reliable website, such as Amazon, Home Depot, etc. Once you find the unit price, you can multiply the number of items by the unit price to get the overall cost.

To find the number of trash cans or benches needed you can find the perimeter of the section, then divide it by how many feet you want each trash can or bench to be apart. We thought 80 ft is a good enough spacing for benches and trash cans. If you get a decimal, then you round it to the nearest whole number. Let's say m_1 is the number of trash cans we would need for the small dog section.

$$m_1 = S \div 80$$

80 ft is how many feet apart we want each trash can to be. For the small dog's perimeter, we would only include the outside fencing, not the fencing inside the shape.

In the number of items, you would have some numbers that would stay the same independent of your space like the number of fountains with a dog shower and vending machines.

5. Conclusions

5.1 - Analysis of Our Project

Strengths

During the development of our plan, we listed two different types of strength that our park has: Mathematical strengths and model strengths. The difference between the two is that mathematical strengths are strengths that have to do with the shapes and



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adjustments to the shapes in the park, and model strengths are strengths that are applied by the things we add inside the shape of the park. The mathematical strengths we see are we have a formula for our shape that can be easily scaled up and down, we based the amount of land provided for the small dog area on percent distribution, and we avoided corners in our shape, so dogs do not get cornered by other dogs. The model strengths we have found are we have a vending machine dedicated to dog treats, we have shade for people alike, we have sitting and eating areas around the park, and we have features to try and make every dog and owner happy.

Limitations

Just like there is strength in everything there are some flaws that are unavoidable due to our park design. These include that we are not able to use the maximum space of the section due to our shape design, The small dogs only get 40% of the dog park area and our vending area takes a fair amount of space from the All-dogs area, and the grass in the play area will get muddy if it rains or snows.

6. Concluding Paragraph

We designed a dog park, composed of two semi-circles and a square. We used the maximum space possible and located our dog park to a location convenient for our visitors. We provided a separate space for small dogs, and we have concession area for our visitors to socialize. Our solution is very effective, because it answers all the questions. These questions state that we must create a dog park that fits everyone's needs, especially our visitors, and is a dog park any owner would like to go to. Said questions include the space our dog park would require, the cost, and the planning for our dog park. Even if any of these variables changed, this would still be a very enjoyable park. If the Director of Parks & Recreation provided us a different amount of space, we would proportionality increase or decrease the area we are provided. If they provided us with a different amount of money, we would get rid of some of our negotiable items or add some items. Our cost would vary greatly given these negotiable items. Due to wanting minimal cost, we tried to minimize fencing with a maximum area. To make this park even more enjoyable for the visitors and their dogs, we have shade, and fun playsets to enhance the experience of our park's visitors.



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7. Letter

Dear Director of Middleville Parks and Creations,

You should choose our park because our dog plan provides good enough space for small dogs and all dogs, and dog owners will like what our park has to offer. Firstly, we made a small dog area where small dogs can have their own little “safe space” to play. This takes approximately 41% of our dog park. We have an *all-dogs* area in which all kinds of dogs can play and socialize. That takes about 47 % of the dog park. Then we have concessions. This includes restrooms, tables, and vending machines. The all-dog area is the largest of the three areas to ensure there is enough room for the large amounts of dogs entering that area to have room to play. These areas have a very reasonable price for the items we chose. We had to make sure that we had all the items that are necessary for our park. These are our non-negotiables. We also have the negotiables, which include picnic tables and playsets. This means the budget has flexibility so that if any of our variables change, we can easily adjust.

We chose the area we did for the dog park because it met all the criteria, we set for the park away from the skate park, does not interfere with anything other than an open field, and has enough space to upscale and downscale a good-sized park. The space that we chose also includes a small forest for shade. The shape we chose also fits this area as well as provides adequate space for the dogs to play.

We chose this shape for our dog park because it doesn't have a lot of corners, so dogs don't get cornered and bullied, but also because it is easy to separate small dogs and big dogs, leaving them with enough space to play.

One advantage of our park is that it generates revenue which goes towards maintaining the park. We have vending machines set up so everyone can have their snacks. One of the machines is for food like chips and sandwiches, another is for dog treats, and the final one is for drinks. Because many owners care about their dogs and like to treat their dogs, this will be a system based on the consumers' wants to earn a steady stream of cash.

As we went through the thinking process, we decided to propose certain rules and fines for rule-breakers to ensure the well-being of the citizens of our park. For example, in the concession area, every dog is required to be on a leash so that everyone is safe while they're eating. We put fines in place along with these rules to better enforce them.

In conclusion, you should pick our park because of the many ways we try to ensure everyone has fun at our park. We have created the best experience possible for our purpose. This Park is always clean because of our team of relentless janitors and employees working to keep their best to keep this park clean. We have an amazing system of regularly spaced seating in case you are tired or just want to watch your dog enjoy itself.

Sincerely,

Team 12263



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9. Association of Professional Dog Trainers
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10. Other Links; Products and Materials
[Vending Machines](#)
[Double Gates](#)
[All Purpose Fountain with two pet bowls](#)
[Fountain with Shower](#)
[Benches](#)
[Fences](#)
[Gates](#)



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